Question 2: Differentiation and Integration.

a) average rate of change use $m = \frac{y_2 - y_1}{x_2 - x_1}$

$y = 4x^2 - x + 2 \qquad x_1 = 0 \quad x_2 = 2$.

when $x_1 = 0$: $y_1 = 4(0)^2 - (0) + 2 = 2$.

$x_1 \; y_1$
$(0, 2)$

when $x_2 = 2$: $y_2 = 4(2)^2 - (2) + 2 = 16$

$x_2 \; y_2$
$(2, 16)$

$\therefore$ average rate of change $(m) = \frac{16 - 2}{2 - 0} = \frac{14}{2} = 7$

b) $y = x^3 + 3x^2 - 9x + 5$.

i) Find $dy/dx$ (First derivative) and $d^2y/dx^2$ (second derivative)

use the power Rule of differen

$dy/dx = 3x^2 + 6x - 9$

$d^2y/dx^2 = 6x + 6$.

→over

ii) coordinates of the max and min turning points.

let $\frac{dy}{dx} = 0$ and solve for $x$.

$\therefore 3x^2 + 6x - 9 = 0$

$(3x - 3)(x + 3) = 0$ by factorisation.

$3x - 3 = 0 \qquad x + 3 = 0$

$3x = 3 \qquad x = -3$

$x = 1$

therefore, have turning points where $x = 1$ and $x = -3$

$d^2y/dx^2 = 6x + 6$.

$x = 1 : 6(1) + 6 = 12$ (positive answer) Minimum point.

$x = -3 : 6(-3) + 6 = -12$ (negative answer) Maximum turning point.

sub $x = 1$ and $x = -3$ back into the original equation to get the corresponding $y$ values and the coordinates of the max and min points.

when $x = 1$, $y = (1)^3 + 3(1)^2 - 9(1) + 5$

$= 0$.

$\therefore (1, 0) \rightsquigarrow$ minimum turning point

when $x = -3$, $y = -3^3 + 3(-3)^2 - 9(-3) + 5$

$= 32$

$\therefore (-3, 32) \rightsquigarrow$ maximum turning point.

c) $y = \underbrace{(3x^2+x)}_{u}\underbrace{(4x-5)}_{v}$ Product Rule.

$\frac{df}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$

in formula sheet.

$u = 3x^2+x$, $du/dx = 6x+1$

$v = 4x-5$, $dv/dx = 4$

sub into the formula:

$$\frac{df}{dx} = (3x^2+x)(4) + (4x-5)(6x+1)$$
$$= 12x^2+4x+24x^2+4x-30x-5$$
$$= 36x^2-22x-5$$

d) $h(t) = -3t^2+24t$

i) height after 1.5 seconds:

$$h(1.5) = -3(1.5)^2+24(1.5)$$
$$= 29.25 \text{ m}$$

ii) rate ball is rising after 2 seconds

get the derivative $dh/dt$

$dh/dt = -6t+24$ @ $t = 2$ seconds

then $-6(2)+24$

$= 12$ m/s

→ over

iii) time at which ball reaches max height.

i.e. $\frac{dh}{dt} = 0$ and solve for $t$.
let

$$-6t + 24 = 0$$
$$6t = 24$$
$$t = \frac{24}{6}$$

$t = 4$ seconds.

e) area enclosed by the curve $y = 2x^2 + 2x + 2$ and $x = 1$ and $x = 4$

Definite Integral.

$$\int_1^4 (2x^2 + 2x + 2)\, dx$$

$$= \left[ \frac{2x^3}{3} + \frac{2x^2}{2} + 2x \right]_1^4$$

$$= \left[ \frac{2x^3}{3} + x^2 + 2x \right]_1^4$$

sub in the upper bound for $x$ ↙ ↘ sub in the lower bound for $x$

$$= \left[ \frac{2(4)^3}{3} + (4)^2 + 2(4) \right] - \left[ \frac{2(1)^3}{3} + (1)^2 + 2(1) \right]$$

$= 63$.